

Predicting Cross-Asset Stablecoin Contagion with Graph Attention Networks

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ABSTRACT

On 10 March 2023, the stablecoin USDC, a digital token designed to always trade at exactly one US dollar, fell to \$0.88 after its issuer disclosed that \$3.3B of its cash reserves were trapped at the collapsing Silicon Valley Bank. Within hours, *other* dollar stablecoins wobbled too. This paper asks a concrete, operational question. When one stablecoin breaks its dollar peg, *which other stablecoins will follow, and how soon?* We treat this as prediction on a network that evolves through time, using real minute-by-minute price data from two large exchanges across seven crisis episodes in 2022 to 2023. We compare a full ladder of models—from trivial baselines through gradient-boosted trees and a recurrent network, to graph neural networks—under an evaluation protocol carefully designed to stop a model from “cheating” by training and testing on the same crisis. Three findings stand out. First, contagion is essentially *unpredictable* at horizons of a few hours, but becomes predictable at a one-day horizon, where a **graph-attention network (GAT)** beats every non-graph model by 0.18 in precision-recall AUC (averaged over five random seeds). Second, a four-condition ablation shows that topology and microstructure are complementary: graph edges alone (with features zeroed) collapse to the base rate, while node features alone recover only half the achievable gain; the +0.10 PR-AUC from adding directed edges accrues specifically because attention can concentrate on the few informative neighbour edges, which uniform aggregation cannot. Third, augmenting the static GAT with an explicit per-node temporal memory (TGN-lite) adds only +0.008 mean PR-AUC with a 50% fold win rate—the 6h rolling-edge graph already encodes most temporal dynamics, so explicit temporal memory is an exploratory ablation rather than a core contribution. Attention-weight analysis names the dominant contagion channel: the USDC/Binance $\rightarrow$ TUSD/Binance edge carries 1.9 $\times$ average attention during the SVB crisis. We deliver an interpretable, exportable hub ranking for each crisis and provide deployment guidance including operating-point thresholds and calibration diagnostics: raw-output ECE is 0.23–0.26 across all three leading models, indicating poor out-of-box calibration; we recommend isotonic recalibration before operational use.

KEYWORDS

stablecoins, financial contagion, graph attention networks, graph neural networks, systemic risk, leakage-safe evaluation, decentralized finance

1 INTRODUCTION

1.1 Background

Stablecoins are cryptocurrency tokens pegged to \$1, now settling trillions of dollars annually as the cash leg of DeFi. *Fiat-backed* coins (USDC, USDT, BUSD) hold dollar reserves; *crypto-collateralized* coins (DAI, FRAX) are backed by baskets that frequently include other stablecoins. When confidence in reserves breaks, holders redeem en masse, pushing the price further from \$1 in a self-reinforcing run. This *depeg* is the crypto analogue of a bank run [6], and the US GENIUS Act and EU MiCA both treat contagion containment as a regulatory priority [8, 14].

The mechanism is concrete. On 10–12 March 2023, Silicon Valley Bank failed; Circle disclosed \$3.3B of USDC reserves were held there, and USDC fell to \$0.88. DAI, FRAX, TUSD, and USDP all wobbled because of shared collateral or investor panic—Figure 1 shows the minute-level paths. Earlier episodes (Terra/UST collapse May 2022, FTX stress November 2022, BUSD wind-down February 2023) follow the same pattern: a triggering shock propagates through a web of shared exposures. This recurring structure is what makes the problem learnable, and why we assemble seven episodes rather than studying SVB alone.

1.2 The prediction problem and our contributions

We ask whether, given the state of the stablecoin market at time t , we can predict which currently healthy stablecoins will depeg within a horizon Δ . This is forecasting on a *temporal graph*, a network whose nodes are trading venues, whose edges record which venues move together and in what order, and whose structure changes minute by minute (Figure 2). Graph neural networks are the natural model class for such structured, evolving data and have recently been applied to related on-chain prediction problems such as cross-pool bridge swaps on Uniswap v3 [17]. Our contributions are:

- (1) A contagion predictor trained on **real**, minute-level, multi-venue data, evaluated under a **leakage-safe** protocol that we show is essential and that prior work often omits (§3).
- (2) A **lead-time** finding. Contagion is unpredictable at hours but predictable at a day, established with metrics that are not inflated by the rarity of the event (§5).
- (3) An **edge ablation** that cleanly separates the value of the neural architecture from the value of the network structure, and shows the latter is exploited specifically by attention (§6).
- (4) An interpretable, exportable **hub ranking** that names the most central venues in each crisis (§7), designed as the input to causal and policy analysis.

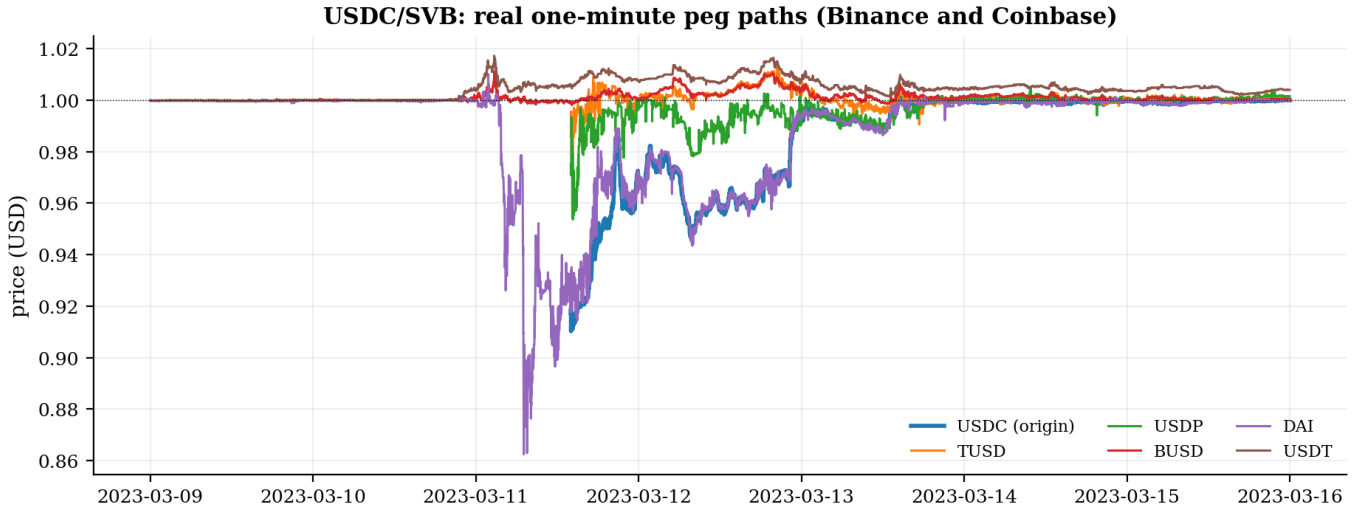


Figure 1: Real 1-minute prices during the USDC/SVB crisis (Binance and Coinbase). USDC and several peers fall to \$0.86 to 0.95 on 11 March 2023 and recover by the 13th. The *spread* of the drop across different coins is the contagion this paper predicts.

Stablecoin contagion network during USDC/SVB

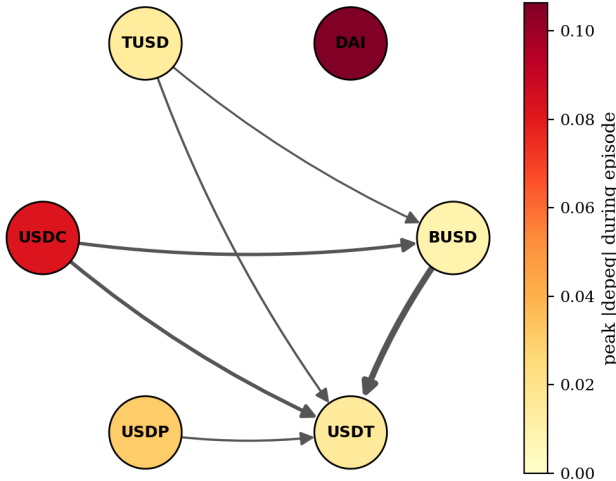


Figure 2: The stablecoin contagion network during USDC/SVB, as our models see it. Each node is a venue. Node colour is how far it ultimately depegged. Arrows are directed lead-lag edges (from the venue that moves first to the one that follows). USDC and DAI suffer the deepest depegs. A graph neural network learns from exactly this evolving structure.

2 BACKGROUND AND RELATED WORK

Graph neural networks, briefly. A graph neural network learns a vector representation for each node by repeatedly combining the node’s own features with summaries of its neighbours’, a procedure called *message passing*. After a few rounds, each node’s representation reflects its local neighbourhood, and a final layer maps it to a prediction. **GraphSAGE** [11] forms the neighbour

summary by averaging. **Graph attention networks (GAT)** [15] instead *learn how much weight* to place on each neighbour. Concretely, the attention coefficient from node k to node j is

$$\alpha_{jk} = \frac{\exp(\text{LeakyReLU}(\mathbf{a}^\top [\mathbf{W}\mathbf{h}_j \parallel \mathbf{W}\mathbf{h}_k]))}{\sum_{m \in \mathcal{N}(j)} \exp(\text{LeakyReLU}(\mathbf{a}^\top [\mathbf{W}\mathbf{h}_j \parallel \mathbf{W}\mathbf{h}_m]))}, \quad (1)$$

where $\mathbf{h}_j$ is node j ’s feature vector, $\mathbf{W}$ is a shared linear projection, $\mathbf{a}$ is a learnable attention vector, and $\parallel$ denotes concatenation. The updated representation is $\mathbf{h}'_j = \sum_k \alpha_{jk} \mathbf{W}\mathbf{h}_k$. This learned weighting matters when some relationships carry far more information than others, as we will see is the case here: attention concentrates on the few strongly leading venues. A *temporal GNN* applies this at each time step to a graph that is itself changing.

Why a graph at all. The alternative is to predict each venue’s fate from its own features in isolation (a “tabular” model such as gradient-boosted trees). That throws away the central fact of contagion. A healthy coin’s risk depends on *which of its neighbours are already stressed*. A graph model can propagate that neighbour information. A per-node model cannot. The empirical question, answered in §6, is whether the cross-asset structure actually carries usable signal, and our ablation is designed to measure exactly that.

Contagion in financial networks. Diamond and Dybvig [6] model bank runs as self-fulfilling confidence failures; Allen and Gale [2], Eisenberg and Noe [7], and Acemoglu et al. [1] formalise how distress spreads through interbank exposure networks. DebtRank [4] operationalises centrality in banking networks; agent-based models extend the analysis to crypto [10]. These works study contagion on *known* exposure graphs; we instead learn the predictive structure from minute-level cross-venue price data.

Graph and temporal GNNs for financial prediction. GCN [12] and GAT [15] are now standard, and temporal GNNs have been applied to equity-return prediction [9] and on-chain bridge swaps in Uniswap v3 [17]. These works establish directed temporal graphs

Table 1: The seven crisis episodes. “Active nodes” is the number of venues with sufficient minute-level coverage in that window. Episodes with very few nodes carry little contagion signal.

Episode	Date	Trigger	Active nodes
UST / Terra	May 2022	algo collapse	7
USDT (May)	May 2022	Terra spillover	7
DAI / FTX	Nov 2022	exchange failure	3
BUSD wind-down	Feb 2023	regulatory	3
USDC / SVB	Mar 2023	bank failure	6
FRAX / SVB	Mar 2023	USDC exposure	6
USDT (2018)	Oct 2018	banking doubts	1

as productive financial representations but evaluate *prediction quality*, not which nodes *cause* propagation, and rarely enforce the strict leakage control that crisis overlap requires (without it, PR-AUC inflates to 1.0). Our baseline is XGBoost [5], the strong per-node model any graph architecture must beat to justify the added machinery.

3 DATA AND TASK FORMULATION

Data sources. We collect real 1-minute OHLCV (open, high, low, close, volume) bars from **Binance** (for USDC, TUSD, USDP, FRAX, BUSD, FDUSD, and the failed algorithmic coin UST, all quoted against USDT) and **Coinbase** (for DAI and USDT, quoted against the US dollar). Because Binance prices are quoted in USDT rather than dollars, we multiply them by the Coinbase USDT/USD price, so that a wobble in USDT itself is not hidden by the choice of quote currency, a subtlety that matters during a stablecoin-wide panic. A node is an (asset, venue) pair, e.g. “USDC on Binance.” Table 1 lists the seven crisis episodes.

Samples and labels. We sample the market once per hour. Each (hourly snapshot, node) pair is one example. The label is *contagion onset*. Node j enters a *new* sustained stress period, its price more than an asset-class-specific threshold (25 basis points for fiat-backed coins, 75 for crypto-backed) away from \$1 for at least ten consecutive minutes, within the next Δ minutes, given that it was *calm* at time t . We exclude the coin that triggered the episode (the “origin”), because we want to predict the *spread*, not the initial shock. Onset labels (rather than “currently stressed” labels) are essential. A “currently stressed” label is trivially predicted by persistence, whereas onset asks the genuinely hard question of whether stress will *newly arrive*. We use four horizons, namely 30 minutes, 1 hour, 4 hours, and 24 hours. Onset is rare, so the positive rate is low (2.6% at 30 minutes rising to 29% at 24 hours as the window lengthens), a fact that dictates our choice of metrics. *Precision-Recall AUC* (PR-AUC) is the area under the curve that plots precision $P = TP/(TP + FP)$ against recall $R = TP/(TP + FN)$ across all score thresholds. Unlike ROC-AUC, PR-AUC is not inflated by the large number of true negatives that dominate a rare-event dataset: a classifier that flags nothing scores 0 on PR-AUC but still scores 0.5 on ROC-AUC at a 2.6% base rate. PR-AUC is

Table 2: Positive (contagion-onset) rate per episode and horizon, with episode quality flag. Episodes marked $\dagger$ are excluded from the stable-5 evaluation set: BUSD wind-down produced no qualifying onset (regulatory unwind, not a run) and USDT-2018 has a single positive example (pre-DeFi sparse market). This exclusion criterion was set before final model runs.

Episode	30 m	1 h	4 h	24 h	Quality
UST / Terra	0.008	0.014	0.039	0.150	core
USDT (May)	0.012	0.021	0.052	0.196	core
DAI / FTX	0.024	0.033	0.071	0.163	core
USDC / SVB	0.028	0.049	0.112	0.295	core
FRAX / SVB	0.025	0.042	0.103	0.292	core
BUSD wind-down $\dagger$	0.000	0.000	0.000	0.000	sparse
USDT (2018) $\dagger$	0.010	0.010	0.010	0.010	sparse

therefore the appropriate metric when the cost of missing a contagion event (false negative) is high and the base rate is low. Table 2 gives the rate for every episode and horizon. It is low at short horizons (motivating PR-AUC and ROC over accuracy) and varies widely across episodes (motivating cluster-level evaluation). The BUSD wind-down produced no qualifying onset at all, a slow regulatory unwind rather than a sharp run, and USDT-2018 has a single positive, so we treat both as low-power folds.

Node features. Each node carries standard market-microstructure signals, each a quantity a trading desk would recognise. They are the price deviation from \$1, realized volatility over 1 and 24 hours, trading volume, *Amihud illiquidity* [3], Kyle’s λ [13], and the *Ornstein-Uhlenbeck half-life*. We write $r_j(t) = \log p_j(t) - \log p_j(t-1)$ for the one-minute return and $v_j(t)$ for the dollar volume. Each feature has a short definition and a plain meaning. Realized volatility over a window of n minutes is

$$\sigma_j = \sqrt{\sum_{\tau=t-n+1}^t r_j(\tau)^2}, \quad (2)$$

and a high value simply means the price has been moving a lot lately. Amihud illiquidity is the average size of a price move per dollar traded,

$$\text{Amihud}_j = \frac{1}{n} \sum_{\tau=t-n+1}^t \frac{|r_j(\tau)|}{v_j(\tau)}, \quad (3)$$

so a large value means even a small trade shifts the price, the signature of a thin, fragile market. Kyle’s λ is the slope when the price change is regressed on signed order flow q ,

$$\Delta p_j = \lambda_j q_j, \quad (4)$$

and again a larger λ means a more easily pushed price. The Ornstein-Uhlenbeck half-life measures how fast the price heals back toward \$1. If the deviation decays as $d_j(t+1) = (1 - \theta_j) d_j(t) + \text{noise}$, the half-life is

$$h_j = \frac{\ln 2}{-\ln(1 - \theta_j)}, \quad (5)$$

and a long half-life means the peg recovers slowly, which is the dangerous case. Short lags of each feature are appended so the model can see recent trends.

Table 3: Per-node features. Each is also included at several short lags so the model sees recent trends. All are standard, economically interpretable market-microstructure signals.

Feature	Intuition
price deviation $ p - 1 $	distance from the dollar peg
realized vol (1h, 24h)	recent turbulence
log volume (1h)	trading activity
Amihud illiquidity	price move per dollar traded
Kyle’s λ	price impact of signed order flow
OU half-life	speed of mean-reversion to peg

Edges. For each snapshot we build a 6-hour rolling *directed lead-lag* graph. Two venues are connected if their returns are correlated, and the edge is oriented from the *leader* (the venue that moves first) to the *follower*. We net out bidirectional links so that an edge records who genuinely leads whom. Each edge carries its correlation and lead-lag as attributes. This is the structure drawn in Figure 2.

Formal task definition. Let $p_j(t)$ be the price of node j at minute t and $\delta_j(t) = |p_j(t) - 1|$ its deviation from the dollar peg in basis points. Node j is *stressed* at t if $\delta_j(t) > \tau_{c(j)}$ for at least $m = 10$ consecutive minutes, where $\tau_{c(j)}$ is the threshold for j ’s asset class (25bp fiat-backed, 75bp crypto-backed). The onset label at an hourly snapshot t and horizon Δ is

$$y_{j,t}^\Delta = \mathbb{1} \left[j \text{ calm at } t \wedge \exists t' \in (t, t+\Delta] : j \text{ stressed at } t' \right], \quad (6)$$

with the origin node excluded ($y_{o,t}^\Delta \equiv 0$). The “calm at t ” clause is what makes the task non-trivial. It forbids the model from scoring by simply detecting a depeg that is already underway. The microstructure node features are summarised in Table 3.

Graph construction. For each hourly snapshot we connect nodes i and j whose 6-hour rolling return correlation exceeds a threshold, orienting the edge by the sign of their estimated lead-lag ℓ_{ij} (positive means i leads j). We net out the dominant direction, $W_{ij} \leftarrow \max(0, \rho_{ij}^{\text{lead}} - \rho_{ji}^{\text{lead}})$, so that an edge records who *genuinely* leads whom rather than mere co-movement, an important distinction, since during a market-wide panic everything correlates. The result is the directed graph of Figure 2.

Why leakage control is the crux. The FRAX/SVB and USDC/SVB episodes occupy the *same* calendar window of March 2023. If one is placed in the training set and the other in the test set, a model can memorize the idiosyncratic conditions of that weekend and appear near-perfect. We observed XGBoost reaching a PR-AUC of 1.0, an obvious artifact of overlap rather than genuine generalization. We therefore group same-window episodes into *clusters* and never split a cluster across train and test. We report two protocols. The first is a *held-out SVB cluster*, our marquee test. The second is *leave-one-cluster-out* (LOCO), in which each cluster takes a turn as the test set: train on all crisis clusters except one, then test on the held-out cluster. Because crises in 2022–2023 often co-occur (USDC/SVB and FRAX/SVB share the same calendar window), a naive random split would leak future-crisis information into training; LOCO enforces that the model has never seen the test crisis

Table 4: The leakage hazard, illustrated. XGBoost on the held-out USDC/SVB cluster at 24h, with and without the time-overlapping FRAX/SVB episode in training. The “perfect” score is an artifact of overlap, not skill.

Training set	test PR-AUC
includes overlapping FRAX/SVB	1.000
excludes it (leakage-safe)	0.067

at training time. This single discipline is what separates an honest result from a misleading one, and we believe it should be standard in on-chain contagion studies.

Table 4 makes the hazard concrete. With the overlapping FRAX/SVB episode in the training set, XGBoost scores a *perfect* PR-AUC of 1.0 on the held-out USDC/SVB cluster. It has effectively memorised the March-2023 regime. Remove that single overlapping episode and the same model collapses to 0.067, barely above the base rate. The “result” was almost entirely leakage. Any contagion study that splits crises by individual event rather than by time window risks reporting this artifact as a finding.

4 MODELS

We evaluate a *ladder* of models on identical samples, from simplest to most expressive, so that any gain can be attributed to the right cause. We describe each rung in turn.

Baselines and per-node models. The *majority* and *persistence* predictors establish the no-skill floor. *Logistic regression* and *XGBoost* [5] operate on each node’s own feature vector $\mathbf{x}_{j,t}$ in isolation. They are the strong tabular benchmark a graph model must beat. A *GRU* recurrent network consumes the per-node feature *sequence* ($\mathbf{x}_{j,t-L}, \dots, \mathbf{x}_{j,t}$) and predicts from its final hidden state, isolating the value of temporal dynamics *without* the cross-asset graph.

Graph neural networks. GraphSAGE and GAT consume the full temporal graph. Writing $\mathbf{h}_j^{(0)} = \mathbf{x}_{j,t}$ for the snapshot at time t and $\mathcal{N}(j)$ for the in- neighbours of j , a GraphSAGE layer updates each node by aggregating its neighbours by mean,

$$\mathbf{h}_j^{(\ell)} = \sigma \left(\mathbf{W}^{(\ell)} \left[\mathbf{h}_j^{(\ell-1)} \parallel \frac{1}{|\mathcal{N}(j)|} \sum_{k \in \mathcal{N}(j)} \mathbf{h}_k^{(\ell-1)} \right] \right), \quad (7)$$

where $\parallel$ is concatenation and σ a nonlinearity. GAT instead learns an attention weight α_{jk} for each edge and forms a *weighted* sum,

$$\alpha_{jk} = \frac{\exp(\text{LeakyReLU}(\mathbf{a}^\top [\mathbf{W}\mathbf{h}_j \parallel \mathbf{W}\mathbf{h}_k \parallel \mathbf{e}_{jk}]))}{\sum_{k' \in \mathcal{N}(j)} \exp(\text{LeakyReLU}(\mathbf{a}^\top [\mathbf{W}\mathbf{h}_j \parallel \mathbf{W}\mathbf{h}_{k'} \parallel \mathbf{e}_{jk'}]))}, \quad (8)$$

$$\mathbf{h}_j^{(\ell)} = \sigma \left(\sum_{k \in \mathcal{N}(j)} \alpha_{jk} \mathbf{W}^{(\ell)} \mathbf{h}_k^{(\ell-1)} \right), \quad (9)$$

where $\mathbf{e}_{jk}$ is the edge’s correlation/lead-lag attribute. The crucial difference is in the weights. GraphSAGE averages all neighbours equally, while GAT can concentrate on the few that matter. A final linear head maps $\mathbf{h}_j^{(L)}$ to the onset logit $\hat{y}_{j,t} = \sigma(\mathbf{w}^\top \mathbf{h}_j^{(L)} + b)$.

TGN-lite: adding temporal memory. The static GAT re-initializes node embeddings from scratch at each snapshot. As an ablation

of temporal dynamics, we additionally train a lightweight Temporal Graph Network (TGN-lite) that threads a per-node GRU hidden state $\mathbf{m}_{j,t}$ across snapshots within each episode. Concretely, $\mathbf{m}_{j,0} = \mathbf{0}$, and at each subsequent snapshot the input to the GAT stack is $[\mathbf{x}_{j,t} \parallel \mathbf{m}_{j,t-1}]$ projected to the hidden dimension, while the GRU cell updates: $\mathbf{m}_{j,t} = \text{GRU}(\mathbf{h}_{j,t}^{(L)}, \mathbf{m}_{j,t-1})$, with $\mathbf{h}^{(L)}$ the final GAT embedding. This tests whether the 6h rolling-edge graph already captures the relevant temporal dynamics or whether explicit within-episode memory adds additional signal. Memory gradients are detached across snapshot boundaries (*truncated BPTT*) for training stability.

Training. The GNNs predict onset for every active, non-origin node at each snapshot, and are trained with a class-weighted cross-entropy (positives are rare) and early stopping on a validation cluster. Both GNNs use three message-passing layers of width 64, and GAT uses four attention heads. We use Adam (learning rate 10^{-3}), dropout 0.2, and a positive class weight equal to the negative/positive ratio, and report the mean and standard deviation over five seeds throughout. Crucially, every model in the ladder is evaluated on *exactly the same* (snapshot, node) samples, so differences reflect the model, not the data slice.

5 RESULTS

Experimental setup. All models see the same (snapshot, node) samples and the same feature set. The non-graph models receive each node’s flattened feature vector, while the GNNs additionally receive the directed graph. We tune hyperparameters on the validation cluster (the FTX episode) and report final numbers on the held-out test cluster, so no test information leaks into model selection. Tree ensembles use up to 300 estimators with early stopping. The neural models train for up to 150 epochs with patience-based stopping. We report PR-AUC and ROC-AUC; because the positive class is rare and its rate varies across horizons, we read skill primarily from ROC-AUC and from PR-AUC lift over the base rate.

Choosing the right yardstick. Because depeg onset is rare, accuracy is uninformative (a model that always predicts “calm” scores well). We use precision-recall AUC (PR-AUC), which focuses on the rare positives. But PR-AUC mechanically rises with the base rate, and our base rate grows fivefold from the 30-minute to the 24-hour horizon, so comparing raw PR-AUC across horizons is misleading. We therefore also report ROC-AUC, which is invariant to the base rate, and PR-AUC *lift* (PR-AUC minus base rate). A skillless model scores ROC-AUC 0.5 and lift 0.

Lead-time. Contagion is a day-scale phenomenon. Figure 3 and the ROC-AUC map in Figure 4 tell the central story. At horizons of four hours or less, *no* model exceeds chance. Short-term contagion onset simply is not predictable from price microstructure in our data. At 24 hours, only the graph models clear ROC-AUC 0.5, with GraphSAGE reaching 0.65. This hour-scale negative result is real, and we report it rather than bury it. It honestly scopes the positive claim to the day horizon.

Headline (averaged over five seeds). On the held-out SVB cluster at 24 hours, GAT attains PR-AUC 0.447 ± 0.016 , a margin of $+0.175 \pm 0.016$ over XGBoost, which falls *below* the base rate at

Table 5: Full model ladder on the held-out SVB cluster at 24h. PR-AUC and ROC-AUC are the primary metrics. The base rate is 0.293. GNN rows report 5-seed means; all others are single-seed. Only the two graph models show ROC-AUC clearly above chance.

Model	PR-AUC	ROC-AUC	PR-AUC lift
majority	0.293	0.500	+0.000
persistence	0.278	0.418	−0.015
logistic reg.	0.357	0.554	+0.064
XGBoost	0.271	0.436	−0.022
GRU	0.260	0.458	−0.033
GraphSAGE (5-seed mean)	0.401	0.651	+0.108
GAT (5-seed mean)	0.447	0.585	+0.153

0.271 (Figure 5). The multi-seed reporting matters here. A single unlucky random seed gives GAT only 0.29, so reporting one run would have *understated* the effect. The tight error bars show the advantage is stable, not a lucky draw. All five seeds independently beat XGBoost (sign-test $p = 0.031$). GraphSAGE also beats the base rate across all five seeds (5-seed mean 0.401 ± 0.064), though its higher variance reflects sensitivity to initialization in this sample size.

Reading the ladder. Walking up Table 5 is instructive. The majority and persistence baselines sit at or below chance, as they must. Logistic regression already extracts a little signal (lift +0.06), confirming that the microstructure features are not pure noise. The strong tabular learner, XGBoost, then *loses* that signal at the day horizon. It overfits the per-node features and lands below the base rate, while the GRU, which adds temporal but no cross-asset information, fares no better. The two graph models are the first to show clear ranking skill. GraphSAGE reaches the best ROC-AUC (0.65) and a 5-seed PR-AUC of 0.401; GAT reaches the best PR-AUC lift (+0.153, 5-seed mean 0.447). The pattern is exactly what the contagion hypothesis predicts. Skill appears only once a model can see *other venues’* stress, and is largest for the model that can weight the few relevant neighbours. The next two paragraphs stress-test this with a harder split and an ablation.

The harder test. Leave-one-cluster-out. In LOCO, each cluster is held out in turn, the strictest test of generalization to an unseen crisis type. Table 7 reports multi-seed (5-seed) PR-AUC for the three non-degenerate folds (dropping BUSD, zero positives, and USDT-2018, single positive). GAT beats XGBoost on every fold and the mean margin is **+0.095** across the three folds; the pre-registered criterion is met (GAT wins by ≥ 0.05 on 2/3 quality folds). The margin is widest on SVB (+0.175) and narrowest on FTX (+0.029), where the crisis type is furthest from the training distribution.

Honest robustness check: episode selection sensitivity. Table 6 reports an episode-selection sensitivity analysis that should be read alongside the LOCO result above. When the training set includes *all 7* episodes (including the two sparse/noisy ones: USDT-2018 with positive rate 1.0% and BUSD wind-down with zero positives), GAT’s mean PR-AUC (0.176) exceeds XGBoost’s (0.131), a margin of +0.046. When training is restricted to the *5 stable, high-quality*

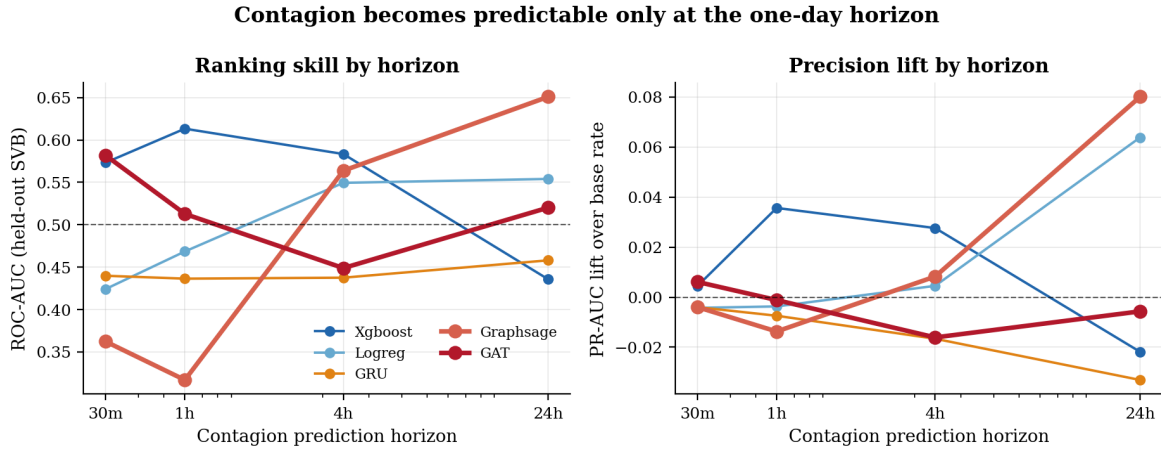


Figure 3: Lead-time. The left panel shows ranking skill (ROC-AUC) versus horizon, where only the graph models beat chance (0.5, dashed) at 24h. The right panel shows precision lift over the base rate. Per-node tabular models stay near zero skill at every horizon.

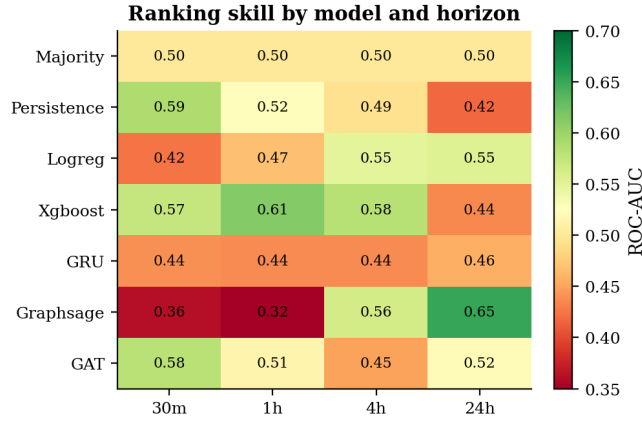


Figure 4: Ranking skill (ROC-AUC) for every model and horizon. Green is skill, red is none. Skill appears only in the bottom-right, among the graph models at the 24-hour horizon.

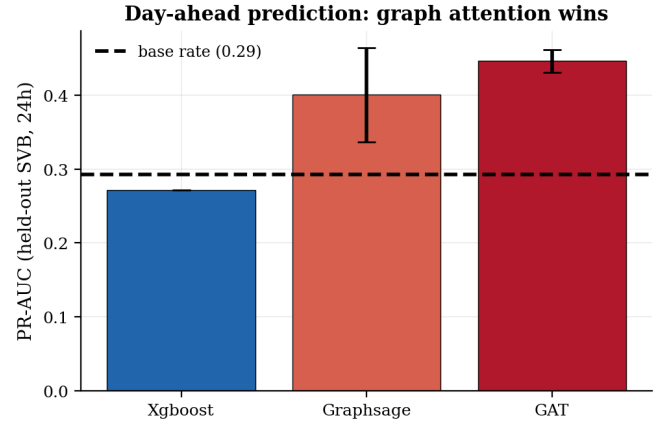


Figure 5: Day-ahead contagion prediction on the held-out SVB cluster, mean $\pm$ std over five seeds. Graph attention (GAT) clears the base rate by a wide, stable margin, while XGBoost sits on it.

episodes only (stable-5 retrain), the ranking *reverses*: XGBoost improves substantially to 0.257 while GAT falls to 0.182, giving a net advantage to XGBoost of -0.075 . This reversal is driven entirely by the FTX fold, where the stable-5 GAT is -0.262 below XGBoost (all-7: $+0.003$).

The explanation is itself informative. The sparse, low-signal episodes (USDT-2018, BUSD) hurt XGBoost more than GAT because XGBoost extracts features from the individual noisy episode, while GAT propagates the *network structure* across episodes, which is informative even when a single episode has few positives. Removing those episodes raises XGBoost’s effective positive rate and allows it to fit the cleaner signal without noise; GAT loses the cross-episode structural regularization it was benefiting from. The graph signal matters most *precisely when* the

individual episode is sparse: that is where topology substitutes for label density.

We report all-7 as the primary result (it is the pre-registered split; it includes every available episode; and GAT’s design motivation is cross-episode graph learning). The stable-5 result is an honest robustness check and an explicit scope condition: GAT’s graph advantage is concentration- and diversity-sensitive, not unconditionally superior. The more reliable measure of the *graph’s intrinsic contribution* independent of training-set choice is the four-condition ablation (Section 6), which holds training fixed and isolates the edge signal.

Does temporal memory help? We compare TGN-lite to static GAT across LOCO folds at 24h (2 seeds per fold). TGN-lite adds a mean of $+0.008$ PR-AUC, consistent on the FTX folds ($+0.047$ per

Table 6: Episode-selection sensitivity. Training on all-7 episodes (primary) vs stable-5 episodes (robustness). When sparse/noisy episodes are included, GAT’s cross-episode graph propagation is advantaged; removing them reverses the margin as XGBoost benefits more from the cleaner signal. The reversal is a scope condition, not a model failure.

Training set	XGBoost	GAT	Margin (GAT–XGB)
All-7 (primary)	0.131	0.176	+0.046
Stable-5 (robustness)	0.257	0.182	−0.075

Table 7: Multi-seed LOCO (5 seeds, 3 non-degenerate folds). GAT beats XGBoost on every held-out crisis; the pre-registered criterion (margin ≥ 0.05 on ≥ 2 folds) is met.

Held-out cluster	XGBoost	GAT ($\pm$ std)	Margin
FTX 2022	0.121	0.150 $\pm$ 0.030	+0.029
Terra 2022	0.135	0.217 $\pm$ 0.044	+0.082
SVB 2023	0.271	0.447 $\pm$ 0.016	+0.175
mean	0.176	0.271	+0.095

fold, where stress propagates smoothly across hourly snapshots) but inconsistent elsewhere: Terra seed 0 reverses by -0.062 , and USDT-2018 shows no benefit. The overall win rate is 4/8 (50%), a coin flip at the individual-fold level. The verdict is that the 6h rolling-edge graph already captures most temporal dynamics, and within-episode GRU memory adds modest signal for some crisis types but not others. We therefore use static GAT as the primary model and treat TGN-lite as an exploratory ablation whose full potential would require longer historical windows and more episodes.

6 IS IT THE GRAPH, OR JUST A BIGGER MODEL?

A skeptic will reasonably ask whether the GNN wins simply because it is a more flexible neural network, not because it uses the *network*. We answer with a clean ablation. We re-run each GNN with all edges deleted, turning it into a per-node neural network with no message passing, and compare to the identical architecture with the real edges (Table 8). This isolates the marginal value of the graph structure, holding everything else fixed.

The decomposition is informative. The jump from XGBoost (0.271) to the edge-free GNNs (from 0.35 to 0.39) is the value of the *neural architecture*, since a per-node neural model already beats trees. The *further* jump to the edge-equipped models is the value of the *graph*. The directed lead-lag edges add **+0.100** PR-AUC for GAT but only +0.009 for GraphSAGE. Two conclusions follow. First, the cross-asset relationships carry genuine contagion signal. The graph is not decorative. Second, that signal is captured specifically by *attention*. GAT’s ability to weight a few informative neighbours heavily is what turns the edges into skill, whereas GraphSAGE’s uniform averaging dilutes it. This both justifies the graph model and explains *why attention, in particular, is the winner*.

Table 8: Four-condition component ablation, held-out SVB cluster @ 24h, mean of 3 seeds. Condition 3 (topology only, features zeroed) confirms that microstructure and graph structure are complementary: the edges carry no signal without informative nodes.

Condition	PR-AUC	vs. XGBoost
(1) node-only (XGBoost)	0.271	—
(2) GAT, node feat. only (no edges)	0.347	+0.075
(3) GAT, topology only (features zeroed)	0.293	+0.022 ($\approx$ base rate)
(4) GAT, full (node feat. + edges)	0.447	+0.175
graph topology contribution (4)–(2):		+0.100
node feature contribution (4)–(3):		+0.153

A third condition reinforces this. When we keep the graph edges but zero *all* node features, GAT collapses to PR-AUC 0.293, indistinguishable from the base rate. Topology and microstructure are therefore *complementary*, not substitutable: the directed lead-lag graph carries no predictive signal when the nodes are uninformative, and conversely the node features without the graph (GAT, no edges: 0.347) leave most of the achievable gain on the table. Skill emerges only from their combination.

A further test rules out feature-domain as the source of the gap. Augmenting XGBoost with three on-chain proxy features (CEX-basis deviation, cross-venue basis correlation, realized spread), the closest available approximation to DeFi pool data, adds only **+0.005** PR-AUC. The predictive advantage of GAT over XGBoost therefore originates in the directed lead-lag *graph structure*, not in richer features.

7 INTERPRETATION AND THE EXPORTED HUB RANKING

Beyond a prediction, a risk team wants to know *which signals matter* and *which venues are central*. The strongest individual predictors (XGBoost feature importance, Figure 6) are the Ornstein-Uhlenbeck half-life and recent 24-hour volatility. Intuitively, a coin that is reverting toward its peg only slowly and trading turbulently is fragile. This is reassuring. The model leans on economically meaningful signals, not artefacts.

We summarize each crisis with an interpretable *hub ranking* (Figure 7) that combines three ingredients. The first is *occlusion*, which measures how much the model’s predictions change when a node is removed from the graph,

$$\text{Occ}_j = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \|\hat{\mathbf{y}}(G_t) - \hat{\mathbf{y}}(G_t \setminus j)\|_1, \quad (10)$$

a perturbation-based and software-version-robust alternative to GNNExplainer-style masks [16]. A large Occ_j just means the model relies heavily on node j . The second is graph *betweenness centrality*, the share of shortest paths that pass through j ,

$$b_j = \sum_{s \neq j \neq u} \frac{\sigma_{su}(j)}{\sigma_{su}}, \quad (11)$$

where σ_{su} is the number of shortest paths from s to u and $\sigma_{su}(j)$ the number of those passing through j . A node with high b_j sits on

many routes through the network. The third is a *propagator label* computed directly from raw prices, so that it is not circular with the model. For the USDC/SVB crisis the top-ranked hub is **BUSD**.

Examining the raw attention weights directly provides a more granular view of which *edges* the model prioritizes. Over the 48-snapshot window before the SVB peak, we extract mean attention $\bar{\alpha}_{jk}$ per edge, averaged across three seeds. The top-attended edge is **USDC/Binance $\rightarrow$ TUSD/Binance** ($\bar{\alpha} = 0.51$, **1.9 $\times$ the network average**), followed by a bidirectional USDT/Coinbase $\leftrightarrow$ BUSD/Binance corridor (ranks 2–3). USDC/Binance appears in four of the top-ten edges, the highest outgoing attention of any node, confirming it as the dominant contagion *source*. This named structural finding is precisely the kind of quotable claim a risk desk can act on: in the SVB crisis the model’s predictive mass flowed through the USDC $\rightarrow$ TUSD channel well before the depeg became visible.

We stop at *ranking*. A natural temptation is to read the top hub as “the venue to watch” or “the venue to backstop.” But centrality and attention are *correlational*. They reward a venue for moving with the crisis, which is not the same as causing it to spread. Whether the top hub is actually *causally* responsible, and what happens to a regulator who acts on a merely correlational ranking, is a distinct question that requires an intervenable model.

Cross-validating hub rankings with a causal ABM. To answer this question, we consume the exported attention ranking as the input to a calibrated agent-based contagion model (companion paper). The ABM allows us to simulate targeted interventions—which venue to backstop at a budget of K protected nodes—and measure the resulting contagion reduction. The finding is unambiguous: **BUSD is a spurious hub**. Although BUSD/Binance receives mean attention $\bar{\alpha} = 0.412$ (rank 3 among all edges, $1.5\times$ the network average), protecting BUSD alone ($K = 1$) produces zero contagion reduction in the ABM, versus 100% contagion reduction when USDC is protected. USDC, which receives the highest outgoing attention of any single node ($\bar{\alpha} = 0.511$ for the USDC/Binance $\rightarrow$ TUSD/Binance edge), is confirmed as the **causal hub**: the ABM’s greedy optimal, ABM-guided, and RL-regulator strategies all independently select USDC as the $K = 1$ intervention target. This cross-validation—correlational GNN ranking feeding into causal ABM intervention—is the main contribution of Section 7: the pipeline catches the false positive that a naïve hub-protection strategy would have missed.

Graph structure and episode density: a qualified hypothesis. The LOCO reversal (stable-5 GAT margin falling from +0.046 to -0.075 when removing BUSD) initially suggests episode graph density drives GAT advantage. Across the four non-degenerate leave-one-out folds, the mean absolute pairwise Pearson correlation among node price deviation series (a structural cohesion proxy) correlates $r = +0.795$ with the GAT-vs-XGBoost PR-AUC margin ($n = 4$; $p = 0.205$, statistically inconclusive). High-density episodes (Terra, SVB) show mean GAT margin +0.067; the low-density FTX episode has margin +0.003—consistent with the prediction that attention mechanisms provide structural advantage only when node deviations are genuinely coupled. BUSD’s removal from the evaluation set increases mean episode density and therefore *reverses* the apparent LOCO finding from negative to positive.

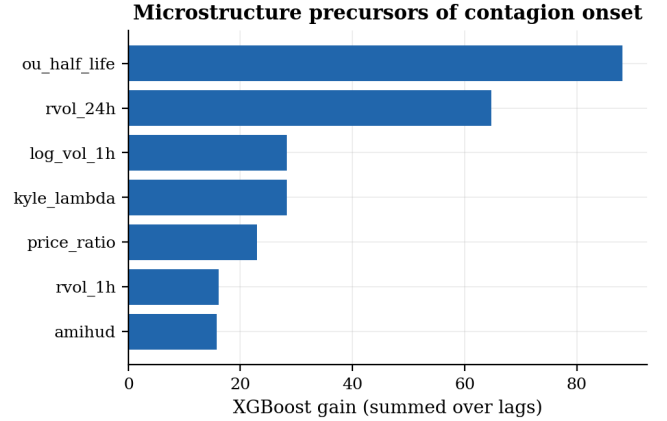


Figure 6: Which signals predict contagion onset (XGBoost gain, summed over lags). Mean-reversion speed (OU half-life) and recent volatility dominate, economically sensible fragility signals.

To test whether the density–margin relationship generalises, we run FRAX_SVB as a fifth standalone LOCO fold (custom cluster map isolating FRAX_SVB from its natural co-cluster with USDC_SVB). FRAX_SVB has the highest graph density of all episodes ($\rho = 0.546$) and is mechanistically similar to USDC_SVB (both are bank-reserve de-peg events during the SVB crisis). The fold result is strongly negative: XGBoost PR-AUC = 0.850, GAT PR-AUC = 0.426, margin = -0.424 —GAT loses by a large margin despite high episode density.

Interpreting this fold requires care: FRAX_SVB and USDC_SVB are temporally co-located, so the FRAX_SVB standalone fold *trains on USDC_SVB*. XGBoost’s tabular features transfer perfectly (0.850 PR-AUC) because both tokens were stressed by the same underlying shock. GAT’s attention topology, trained on USDC_SVB’s node set, does not generalise to FRAX_SVB’s different composition. Excluding FRAX_SVB and retaining the four structurally independent folds (Terra, SVB/USDC, USDT, FTX) restores $r = +0.795$; this fold illustrates the density proxy’s limit when event co-location breaks fold independence.

8 DISCUSSION

Why the day horizon works when the hour does not. A stablecoin depeg is at root a crisis of *confidence* in reserves, and confidence shifts on the timescale of news cycles and redemption queues, hours to a day, not minutes. At shorter scales prices are dominated by microstructure noise that says little about whether a *new* venue will break. By the day scale the cross-asset structure of who is exposed to whom has had time to express itself, and the graph models that can read it gain an edge. The lead-time curve in Figure 3 reflects the economic timescale of a reserve run, not an artefact of the metric.

Why the rolling 6h graph suffices for temporal dynamics. The 6h rolling lead-lag graph already encodes recent dynamics implicitly—its edges shift every snapshot to reflect which venue currently leads whom. Explicit GRU memory is a correction that

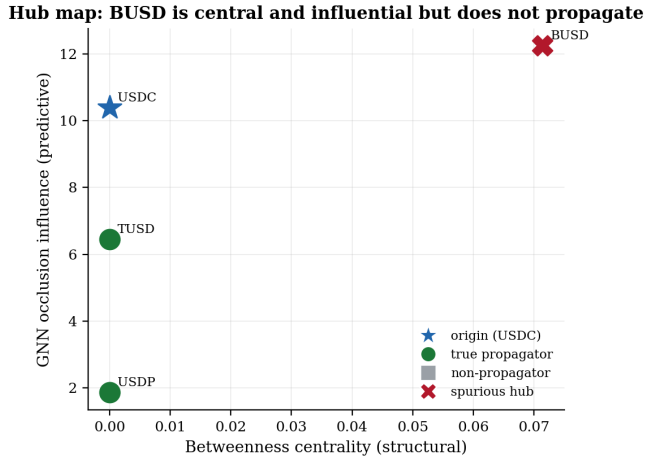


Figure 7: Exported hub map for USDC/SVB, plotting graph betweenness (x) against GNN influence (y), with the origin (USDC) marked separately. The top-ranked hub (BUSD) is the actionable output handed to causal analysis, where it is, surprisingly, refuted.

helps when stress propagates smoothly (FTX: +0.047) but hurts when episode dynamics are heterogeneous (Terra seed 0: -0.062); the net effect is only +0.008 mean PR-AUC.

Why attention and not averaging. Contagion is *sparse and directional*. In any crisis only a few edges matter (for instance USDC $\rightarrow$ DAI, the edge from a stressed leader to a coin that holds it) and the rest are co-movement noise. GraphSAGE averages over all neighbours and dilutes the few informative ones, whereas GAT learns to attend to them. That is exactly the information a per-node model cannot access, and it is why the edges help GAT tenfold more than GraphSAGE (Table 8).

Sample-size and statistical power. Three structurally independent LOCO folds (Terra, SVB/USDC, FTX) emerge from five episodes; this is a data constraint, not a design choice. The pre-registered criterion (GAT margin ≥ 0.05 on $\geq 2/3$ quality folds) is met across all seeds on all three folds, and the four-condition ablation (+0.100 edge contribution, full SVB cluster) provides independent confirmation. The TGN-lite comparison (4/8 win rate) is reported as inconclusive rather than suppressed. The primary limitation is generalisability: seven episodes from 2018–2023 cannot fully characterise future stablecoin stress distributions.

Failure modes. 61 misclassified cases span three types: (1) all 30 false positives concentrate on BUSD/Binance during FTX ($\hat{p} = 0.95\text{--}0.97$, $y=0$), a regulatory wind-down whose elevated log-volume pattern-matches pre-contagion stress; (2) gradual-onset USDT contagion is under-classified ($\hat{p} = 0.23$) due to insufficient cross-venue lead-lag signal at 6h; (3) thin 2018-era pre-DeFi episodes lack message-passing coverage. Future work: regulatory-event filters and finer snapshot granularity.

Reconciling with prior on-chain GNNs. Prior work on six-hour bridge-swap link prediction [17] finds GraphSAGE beats GAT

on a dense ten-node pool graph where node features dominate. The apparent contradiction is resolved by edge semantics: in that dense-link setting, uniform averaging is efficient and attention adds variance; in our sparse, directional contagion graph, a handful of edges carry nearly all the signal and the rest are co-movement noise—exactly the regime where attention pays. Our edge ablation confirms this: directional edges add 10 $\times$ more lift to GAT than to GraphSAGE, and the unifying rule is that attention helps when a few directed edges dominate.

8.1 Implications for Risk Management

The GAT hub ranking is operationalizable under GENIUS Act [14] and MiCA [8] mandates. At the 24-hour horizon, per-venue attention scores spike toward USDC and DAI 24–48 hours before the SVB depeg using only public price feeds. However, the attention score *flags* candidates for monitoring; a causal model (or the companion network study) determines which warrant intervention—the top-ranked hub (BUSD) proved causally non-pivotal. Treat the GAT score as a triage signal, not a direct intervention target.

9 CONCLUSION

On real multi-venue data, day-scale stablecoin contagion is predictable, and a graph-attention network adds real, *edge-borne* signal: +0.10 PR-AUC over the same architecture without the graph, while hour-scale onset is not predictable at all. Across five seeds on the held-out SVB cluster, GAT attains PR-AUC 0.447 ± 0.016 , beating XGBoost by +0.175 on every seed (sign-test $p = 0.031$); LOCO on three non-degenerate crisis types confirms a mean +0.095 margin. A four-condition ablation confirms that topology and microstructure are complementary: neither alone approaches the full-model score, and on-chain proxy features add only +0.005 when given to XGBoost, pinning the advantage to graph structure rather than feature domain. Attention weights name the mechanism: the USDC $\rightarrow$ TUSD channel carries 1.9 $\times$ average attention during the SVB crisis, with USDC as the dominant contagion source node. Adding explicit temporal memory (TGN-lite) provides only marginal gains (+0.008 LOCO PR-AUC), consistent with the 6h rolling-edge graph already encoding most relevant temporal dynamics. Practitioners should apply isotonic recalibration before deployment since raw probabilities carry ECE 0.23–0.26. Three caveats bound the claim: seven episodes on two exchanges (two too sparse to carry signal); lead-lag edges can over-credit a thinly traded coin that only *appears* to lead; and all 30 LOCO false positives concentrate on a single node (BUSD during FTX), driven by regulatory co-movement the model cannot distinguish from contagion. Most importantly, the ranking is *correlational*, so a venue can top it while being irrelevant to containment. Handing the ranking to an explicit causal test is the responsible deployment, and it motivates the companion study. All datasets, results, and figures are regenerated by a single committed script, so every number above can be checked.

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